

Ideation Phase

Empathize & Discover

Date	13 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map helps the team deeply understand users by capturing what they think, feel, see, hear, say, and do. It enables building solutions grounded in real user needs and constraints..

Target User Persona: Small & Marginal Farmer (Rain-dependent region in India)

	THINKS & FEELS • Worried about selling fruits that may be spoiled or rotten. • Concerned about maintaining the quality of fruits for customers. • Hopes for a quick and reliable way to identify rotten fruits.	
SEES • Mixed quality fruits in markets and warehouses. • Some fruits appearing fresh but internally spoiled. • Customers carefully inspecting fruits before buying.		HEARS • Customers complaining about rotten fruits in purchases. • Market vendors discussing fruit quality and losses due to spoilage. • Advice from traders about checking fruits carefully before selling.

	SAYS & DOES • Manually checks fruits before displaying them for sale. • Removes obviously rotten fruits from the basket. • Tries to maintain good quality to keep customer trust.	
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PAINS

- Difficulty in identifying rotten fruits quickly among large batches.
- Financial losses due to spoiled fruits.
- Customer dissatisfaction when rotten fruits are sold accidentally.
- Time-consuming manual inspection process.

GAINS

- Faster and accurate identification of rotten fruits.
- Reduction in food waste and financial loss.
- Better product quality for customers.
- Improved trust and reputation in the market.

How Our Solution Helps

The Smart Sorting – Rotten Fruit Detection using Machine Learning system analyzes fruit images using image processing and deep learning techniques to automatically classify fruits as fresh or rotten. The system enables faster sorting, reduces manual effort, minimizes food waste, and helps vendors and farmers maintain better quality control in the fruit supply chain.